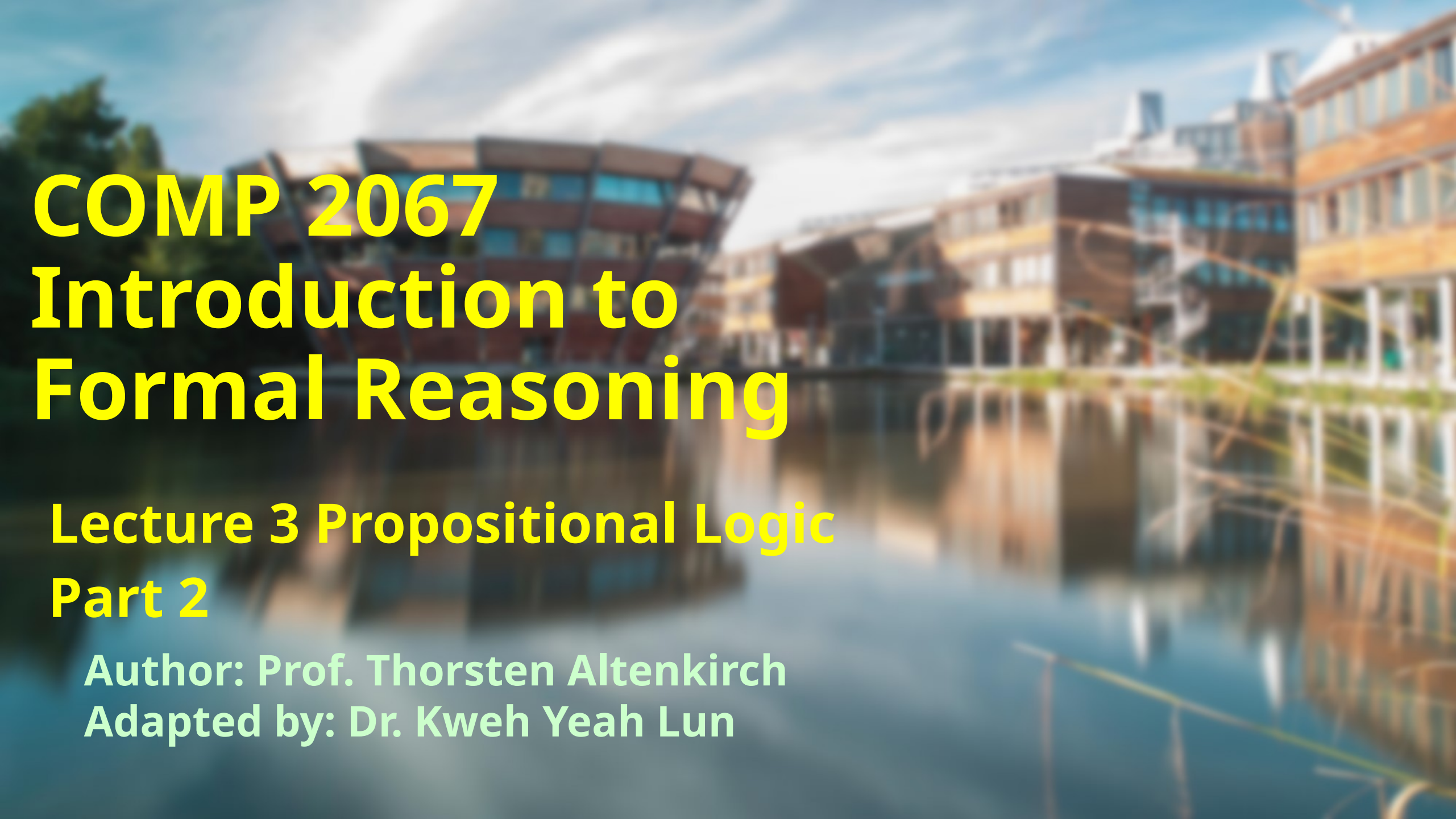




COMP 2067

Introduction to

Formal Reasoning

Lecture 3 Propositional Logic

Part 2

Author: Prof. Thorsten Altenkirch
Adapted by: Dr. Kweh Yeah Lun



Learning Outcomes

At the end of this lecture, you will be able to

- use Lean to prove the **swap theorem**
- explain the concept of **conjunction**
- use tactic **constructor** and **cases** for conjunction
- use Lean to prove the **commutative law** for conjunction
- elaborate and prove the **currying equivalence** theorem
- explain the concept of **disjunction**
- use tactic **left**, **right** and **cases** for disjunction



• Example:

```
theorem swap : (P → Q → R) → (Q → P → R) :=  
begin  
  assume f q p,  
  apply f,  
  exact p,  
  exact q,  
end
```

[Try it!](#)



- After `assume f q p` (which is a short cut for writing three times `assume`) :

```
P Q R : Prop,  
f : P → Q → R,  
q : Q,  
p : P  
⊢ R
```

- Note: `f` represents $P \rightarrow Q \rightarrow R$
- Alternatively: instead of `assume f`, it can be written as `assume pqr`, or `assume p2q2r`



- Now use **f** because its conclusion matches the goal, then:
two goals were left:

```
2 goals
P Q R : Prop,
f : P → Q → R,
q : Q,
p : P
⊢ P
```

```
P Q R : Prop,
f : P → Q → R,
q : Q,
p : P
⊢ Q
```

- After completing the first goal with **exact p** it disappears and only one goal remains:

```
P Q R : Prop,
f : P → Q → R,
q : Q,
p : P
⊢ Q
```

which it can be quickly eliminated using **exact q**.



Conjunction, $\wedge$

- To prove $P \wedge Q$ we need to prove both **P and Q**.
- P and Q can only be true when both P and Q are true.

P	Q	$P \wedge Q$
0	0	0
0	1	0
1	0	0
1	1	1

- In Lean, this is achieved via the **constructor** tactic.

• **Example:**

```
example : P → Q → P ∧ Q :=  
begin  
  assume p q,  
  constructor,  
  exact p,  
  exact q,  
end
```

[Try it!](#)



- **constructor** turns the goal $P \wedge Q$ into two goals:

```
2 goals
P Q : Prop,
p : P,
q : Q
⊢ P
```

```
P Q : Prop,
p : P,
q : Q
⊢ Q
```

```
P Q : Prop,
p : P,
q : Q
⊢ P ∧ Q
```

- then **exact P**, then **exact Q**.



- Example: show that $\wedge$ is **commutative**

Try it!

- After **assume pq**

```
P Q : Prop,  
pq : P ∧ Q  
⊢ Q ∧ P
```

```
theorem comAnd : P ∧ Q → Q ∧ P :=  
begin  
  assume pq,  
  cases pq with p q,  
  constructor,  
  exact q,  
  exact p  
end
```

- **assume pq**: $P \wedge Q$ is the same as assuming both P and Q .
- After **cases pq with p q**, the state

```
P Q : Prop,  
p : P,  
q : Q  
⊢ Q ∧ P
```




- Lean always abstracts the propositional variables we have declared.
- Use `comAnd` with different instantiation to prove the following:

```
theorem comAndIff : P ∧ Q ↔ Q ∧ P :=  
begin  
  constructor,  
  apply comAnd,  
  apply comAnd,  
end
```

Try it!

- In the 2nd use of `comAnd`, Lean instantiate `Q` with `P` and `P` with `Q`.
- Lean will find these instances automatically but in some more complicated examples it may need some help.



The Currying Equivalence

- A statement like $P \rightarrow Q \rightarrow R$ means that R can be proved from assuming both P and Q .
- It is equivalent to $P \wedge Q \rightarrow R$.
- Show this formally by using $\leftrightarrow$.
- Remember that $P \leftrightarrow Q$ is the same as $(P \rightarrow Q) \wedge (Q \rightarrow P)$.
- Hence a goal of the form $P \leftrightarrow Q$ can be turned into two goals $P \rightarrow Q$ and $Q \rightarrow P$ using the `constructor` tactic.



- Example:

```
theorem curry : P ∧ Q → R ↔ P → Q → R :=  
begin  
  constructor,  
  assume pqr p q,  
  apply pqr,  
  constructor,  
  exact p,  
  exact q,  
  assume pqr pq,  
  cases pq with p q,  
  apply pqr,  
  exact p,  
  exact q,  
end
```

Try it!



Disjunction, $\vee$

- To prove a disjunction $P \vee Q$, either prove P or prove Q will be ok.
- This is achieved via the tactics `left` and `right`.
- Examples:

```
example : P → P ∨ Q :=  
begin  
  assume p,  
  left,  
  exact p,  
end
```

```
example : Q → P ∨ Q :=  
begin  
  assume q,  
  right,  
  exact q,  
end
```

P	Q	$P \vee Q$
0	0	0
0	1	1
1	0	1
1	1	1

[Try it!](#)



- To use a disjunction $P \vee Q$, use **cases**
- Example:

```
theorem case_lem : (P → R) → (Q → R) → P ∨ Q → R :=  
begin  
  assume p2r q2r pq,  
  cases pq with p q,  
  apply p2r,  
  exact p,  
  apply q2r,  
  exact q,  
end
```

[Try it!](#)



- After **assume**

```
P Q R : Prop,  
p2r : P → R,  
q2r : Q → R,  
pq : P ∨ Q  
⊢ R
```

- Use **cases pq with p q** which means that use **P ∨ Q**.



- There are **two cases** to consider one for **P** and one for **Q** resulting in two subgoals.

```
2 goals
case or.inl
P Q R : Prop,
p2r : P → R,
q2r : Q → R,
p : P
⊢ R
```

```
case or.inr
P Q R : Prop,
p2r : P → R,
q2r : Q → R,
q : Q
⊢ R
```



- Give yourself a big clap.
- You have survive until here. ◀◀
- Any question?



University of
Nottingham

UK | CHINA | MALAYSIA

Next lecture...

- Logic and algebra



“The greatest teacher, failure is.”

– Yoda. (2017). Star Wars: Episode VIII - The Last Jedi. Lucasfilm.

Hope you all enjoy this lecture!